



HK IMO Training 2022-2023
Problem Set 3
For 8 October 2022



Problem 9. Let k be a positive integer. Consider the set S of all positive integers n satisfying $\lfloor \sqrt{n} \rfloor = k$. Suppose a, b, c, d are elements in S such that $ab = cd$. Prove that $a = c$ or $a = d$.

Problem 10. Let x, y, z be positive real numbers satisfying $x^2 + y^2 + z^2 = 3$. Prove that

$$xyz(x + y + z) + 2021 \geq 2024xyz.$$

Problem 11. Let n be a positive integer. Let $\mathcal{F}$ be a family of some subsets of $\{1, 2, \dots, n\}$ such that the following hold: For any $A, B, C \in \mathcal{F}$, one of $(A - B) \cap C$ and $(B - A) \cap C$ is the empty set. Find the largest possible number of elements in $\mathcal{F}$.

Problem 12. Let $\triangle ABC$ be an acute triangle. The tangents at B and C to the circumcircle of $\triangle ABC$ intersect at P . Consider the variable points X and Y lying on the lines AB and AC respectively, such that P lies inside $\triangle AXY$ and $\angle XPY = 2\angle BAC$. Let Z be the reflection of A in the line XY . Prove that the circumcircle of $\triangle XYZ$ passes through a fixed point when X and Y vary.